SCH4U Lab: pH of Salts and Ka

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Purpose

1. To measure the pH of salt solutions
2. To determine the Ka of weak acids using pH measurements
3. To prepare and test a buffer solution

Introduction

This lab explores the pH of salt solutions, the acid dissociation of weak acids, and the properties of buffer solutions. When a neutralization occurs, a salt is formed. Neutral salts form when a strong acid and a strong base react. A non-neutral salt forms when a weak acid or a weak base is involved, it exhibits either acidic or basic properties due to hydrolysis reactions in water.

Weak acids/bases only partially dissociate in aqueous solution. The strength of a weak acid or bases is quantified by its acid dissociation constant, K_a or K_b respectively. By measuring the pH of buffer solutions containing a weak acid/weak base and its conjugate base/acid, we can determine the K_a/K_b value. A buffer solution resists changes in pH when small amounts of acid or base are added, making it useful in many chemical and biological applications.

This lab consists of three parts. First, the pH of different salt solutions will be measured using a pH probe. Second, buffer solutions will be prepared by combining a weak acid with its conjugate base, and their pH will be recorded to calculate Ka values. Finally, the buffer's ability to resist pH changes will be tested by adding a strong base and comparing the pH change to that of pure water treated the same way.

Safety considerations: Wear gloves at all times during this experiment, as some solutions may cause skin irritation upon direct contact.

Procedure

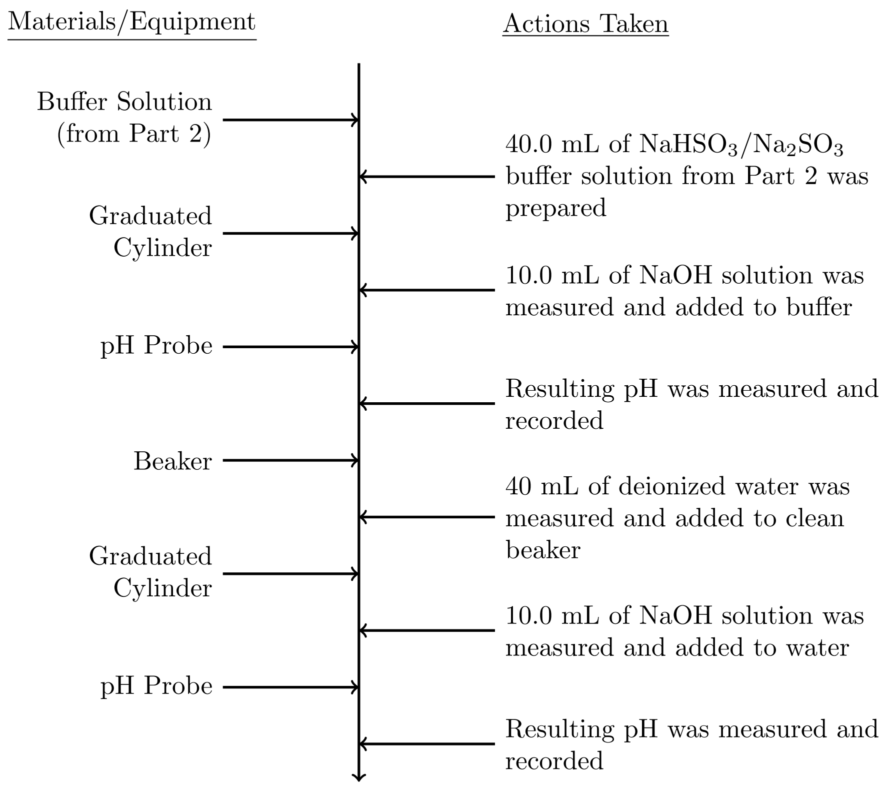
Part 1: pH of Salt Solutions

Materials/Equipment	Actions Taken
Beaker (100 mL)	
	Clean 100 mL beaker was prepared
Graduated Cylinder	
	20.0 mL of 0.100 M NaCl solution was measured and added to beaker
pH Probe	
	pH probe was calibrated according to manufacturer instructions
	pH of NaCl solution was measured and recorded
Deionized Water	
	pH probe was rinsed thoroughly with deionized water
	Procedure was repeated for 0.100 M Na_2CO_3 solution
	Procedure was repeated for 0.100 M $NaHSO_3$ solution
	pH probe was rinsed after each measurement

Part 2: Ka of a Weak Acid

Materials/Equipment	Actions Taken
Beaker (250 mL)	
	Clean 250 mL beaker was prepared
Graduated Cylinder	
	20.0 mL of 0.1 M $NaHSO_3$ solution was measured and added to beaker
	20.0 mL of 0.100 M Na_2SO_3 solution was measured and added to beaker
Glass Rod	
	Solution was stirred thoroughly to mix
pH Probe	
	pH of buffer solution was measured and recorded
Deionized Water	
	pH probe was rinsed thoroughly
	Procedure was repeated using CH_3COOH and $NaCH_3COO$ solutions
	pH of second buffer solution was measured and recorded

Part 3: Buffer Solution Testing



Observations

Part 1: pH of Salt Solutions

Table 1: pH Measurements of Salt Solutions

Solution	**pH**
0.100 M NaCl	
0.100 M Na_2CO_3	
0.100 M $NaHSO_3$	

Part 2: Ka of Weak Acids

Part 3: Buffer Capacity

Table 2: pH of Buffer Solutions

Buffer	**Acid**	**pH**
$NaHSO_3/Na_2SO_3$	HSO_3^-	
$NaCH_3COO/CH_3COOH$	CH_3COOH	

Table 3: pH Changes Upon Addition of NaOH

Solution	**Initial pH**	**After 10 mL NaOH**	**pH Change**
Buffer (HSO_3^-/SO_3^{2-})			
Deionized Water			